

DBMS Test Notes for RAG Chatbot

Purpose

This text-based PDF is designed for testing a Retrieval-Augmented Generation (RAG) pipeline. The text can be extracted directly with PyMuPDF, split into chunks, embedded with Gemini, stored in Qdrant, and retrieved to answer questions.

1. Database Management System

A Database Management System (DBMS) is software used to create, store, organize, retrieve, update, and manage data in a database. It provides an interface between users or applications and the stored data. Common DBMS responsibilities include data definition, query processing, security, integrity enforcement, concurrency control, backup, and recovery.

Advantages of a DBMS

A DBMS reduces unnecessary data duplication, improves data consistency, supports controlled sharing of data, provides access control, and makes backup and recovery easier. It also helps applications remain independent from many physical storage details.

2. Relational Model

In the relational model, data is represented using relations, commonly called tables. A table consists of rows and columns. A row represents a tuple or record, while a column represents an attribute. A relation schema describes the name of the relation and its attributes.

Keys

A super key is a set of one or more attributes that uniquely identifies a tuple. A candidate key is a minimal super key. One candidate key is selected as the primary key. A foreign key is an attribute or set of attributes that refers to a key in another relation and helps maintain referential integrity.

3. Functional Dependency

A functional dependency describes a relationship between attributes. If X functionally determines Y , written $X \rightarrow Y$, then any two tuples that have the same value for X must also have the same value for Y . Functional dependencies are important for identifying redundancy and performing normalization.

Example

Consider `STUDENT(StudentID, StudentName, DepartmentID, DepartmentName)`. If `StudentID` uniquely identifies each student, $\text{StudentID} \rightarrow \text{StudentName}$ and $\text{StudentID} \rightarrow \text{DepartmentID}$ may hold. If each `DepartmentID` identifies exactly one department name, $\text{DepartmentID} \rightarrow \text{DepartmentName}$ also holds.

4. Normalization

Normalization is the process of organizing data in a relational database to reduce redundancy and undesirable dependency while improving data integrity. It usually decomposes a large relation into smaller related relations based on keys and functional dependencies.

Why normalization is needed

Poorly designed tables may repeat the same facts many times. This can waste storage and create inconsistencies. Normalization aims to reduce these problems and avoid insertion, update, and deletion anomalies.

Anomalies

An insertion anomaly occurs when a fact cannot be inserted without adding unrelated data. An update anomaly occurs when the same fact appears in several rows and must be changed in every location. A deletion anomaly occurs when deleting one record unintentionally removes another important fact.

5. First Normal Form (1NF)

A relation is in First Normal Form when every attribute contains atomic values and repeating groups or multivalued fields are removed. Each cell should contain a single value from its domain.

6. Second Normal Form (2NF)

A relation is in Second Normal Form when it is already in First Normal Form and every non-prime attribute is fully functionally dependent on the whole candidate key. Second Normal Form mainly removes partial dependencies that can occur with composite keys.

7. Third Normal Form (3NF)

A relation is in Third Normal Form when it is in Second Normal Form and non-key attributes do not depend transitively on a candidate key. In simple terms, non-key attributes should depend on the key, the whole key, and not on another non-key attribute.

8. Boyce-Codd Normal Form (BCNF)

Boyce-Codd Normal Form is a stronger form of Third Normal Form. A relation is in BCNF if, for every non-trivial functional dependency $X \rightarrow Y$, X is a super key. BCNF addresses certain anomalies that may remain in relations that satisfy 3NF.

Normalization Summary

Normal Form	Main Requirement
1NF	Atomic values; no repeating groups
2NF	1NF plus no partial dependency on a composite candidate key
3NF	2NF plus removal of transitive dependency of non-key attributes
BCNF	Every determinant of a non-trivial dependency is a super key

9. Transactions and ACID Properties

A transaction is a logical unit of database work consisting of one or more operations. Reliable transaction processing is commonly described using the ACID properties: Atomicity, Consistency, Isolation, and Durability.

Atomicity

Atomicity means that all operations of a transaction are completed successfully or none of them are applied.

Consistency

Consistency means a transaction takes the database from one valid state to another valid state while preserving defined constraints.

Isolation

Isolation means concurrently executing transactions should not improperly interfere with one another.

Durability

Durability means that once a transaction is committed, its changes remain stored even after a system failure.

10. SQL

Structured Query Language (SQL) is used to define, manipulate, query, and control relational databases. DDL commands such as CREATE, ALTER, and DROP define database objects. DML commands such as INSERT, UPDATE, and DELETE modify data. SELECT is used to retrieve data. Transaction-control commands include COMMIT and ROLLBACK.

11. Indexing

A database index is an auxiliary data structure that can speed up data retrieval. Indexes improve many search operations but require additional storage and can add overhead to insert, update, and delete operations. B-tree and B+ tree structures are widely used for database indexes.

Suggested RAG Test Questions

Try asking: What is normalization? Why is normalization needed? What are insertion, update, and deletion anomalies? Explain 1NF, 2NF, and 3NF. What is BCNF? What is a functional dependency? What is the difference between a primary key and a foreign key? Explain ACID properties. What is an index in DBMS?